

GEOG5990M Final Assignment

(The cover sheet has been adapted from the word doc. to support use with .ipynb files - please complete all the requested info.)

COURSEWORK COVERSHEET

- Student ID number: 201973991
- Module code: GEOG5990M
- Module title: Programming for Geographical Information Analysis
- Assignment title: Spatial data science report
- Marker: Francesca Pontin
- Declared word count:

Use of Generative Artificial Intelligence (Gen AI) in this assessment – delete one statement as appropriate

- I have made no use of Gen AI
- I have used Gen AI only for the specific purposes outlined in my acknowledgements. Yes

By submitting the work in this notebook you confirm your compliance with the University's definition of Academic Integrity as: "a commitment to good study practices and shared values which ensures that my work is a true expression of my own understanding and ideas, giving credit to others where their work contributes to mine". Double-check that your referencing and use of quotations is consistent with this commitment.

You also confirm that your declared word count accurately reflects the number of words in your submission, excluding the overall title, Gen AI use acknowledgement, bibliography/reference list, text/numbers in tables and figures (although table and figure captions are included in the word count).

Research Question

This project investigates the relationship between socio-economic deprivation and house prices in Leeds.

Specifically, it asks:

Is there an association between Index of Multiple Deprivation (IMD) scores and house prices across LSOAs in Leeds?

This question is important for understanding spatial inequality and can provide insights for policymakers and urban planners.

✓ Data

This project uses three datasets:

- House price data (2021)
- Population data (2021)
- Index of Multiple Deprivation (IMD) data (2019)

All datasets are linked using the LSOA code, allowing spatial comparison across Leeds.

The original house price dataset was large and contained data for all UK regions. To improve efficiency, the dataset was filtered in advance to include only Leeds observations. The cleaned dataset is used directly in this analysis.

```
price_df = pd.read_csv("leeds_house_price_2021(1).csv")

price_df.head()
```

	LSOA_code	house_price	
0	E01011264	207500.0	
1	E01011265	289950.0	
2	E01011266	332500.0	
3	E01011267	228000.0	
4	E01011268	175000.0	

Next steps:

[Generate code with price_df](#)[New interactive sheet](#)

```
pop_df = pd.read_csv("leeds_population_2021.csv")

pop_df.head()
```

```
/tmp/ipykernel_17771/1193157883.py:1: DtypeWarning: Columns (0,1) have mixed types.
pop_df = pd.read_csv("leeds_population_2021.csv")
```

	area_name	LSOA_code	population	
0	Leeds 001A	E01011698	1462.0	
1	Leeds 001B	E01011699	1289.0	
2	Leeds 001C	E01011701	1367.0	
3	Leeds 001D	E01011702	1648.0	
4	Leeds 001E	E01011703	1224.0	

```
merged_df = price_df.merge(pop_df, on="LSOA_code")
```

```
merged_df.head()
```

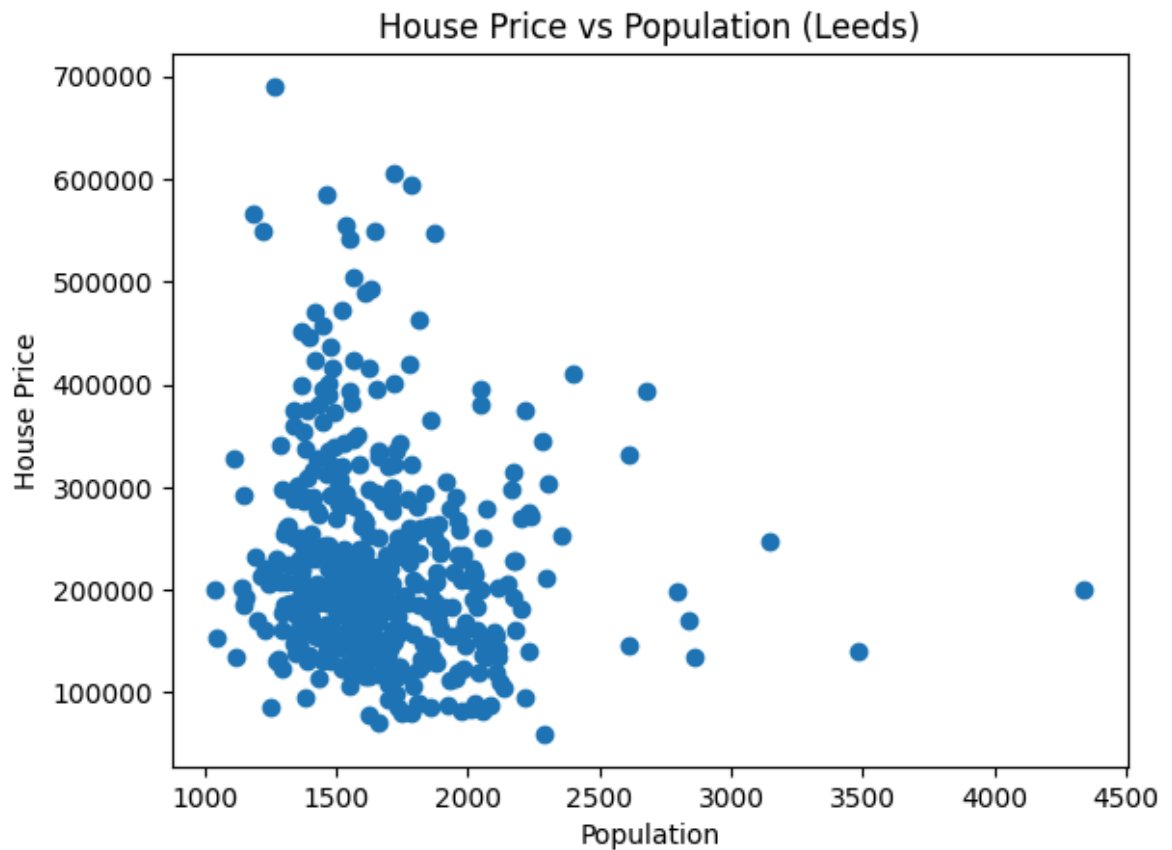
	LSOA_code	house_price	area_name	population	
0	E01011264	207500.0	Leeds 011A	1284.0	
1	E01011265	289950.0	Leeds 009A	1955.0	
2	E01011266	332500.0	Leeds 008A	2613.0	
3	E01011267	228000.0	Leeds 009B	1699.0	
4	E01011268	175000.0	Leeds 010A	1395.0	

Next steps:

[Generate code with merged_df](#)[New interactive sheet](#)

```
import matplotlib.pyplot as plt

plt.scatter(merged_df["population"], merged_df["house_price"])
plt.xlabel("Population")
plt.ylabel("House Price")
plt.title("House Price vs Population (Leeds)")
plt.show()
```



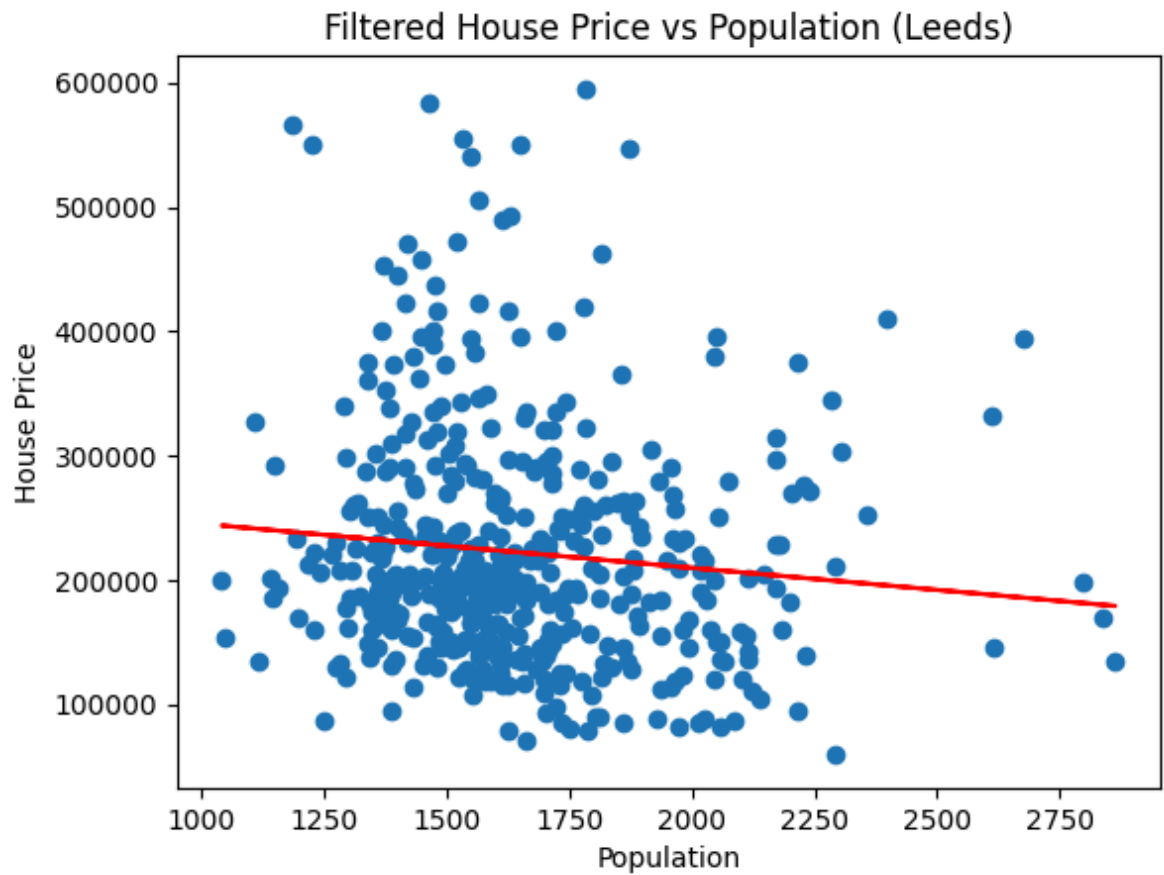
```
#
import numpy as np
clean_df = merged_df[
    (merged_df["house_price"] < 600000) &
    (merged_df["population"] < 3000)
]

#
x = clean_df["population"]
y = clean_df["house_price"]

m, b = np.polyfit(x, y, 1)

plt.scatter(x, y)
plt.plot(x, m*x + b, color='red')

plt.xlabel("Population")
plt.ylabel("House Price")
plt.title("Filtered House Price vs Population (Leeds)")
plt.show()
```



```

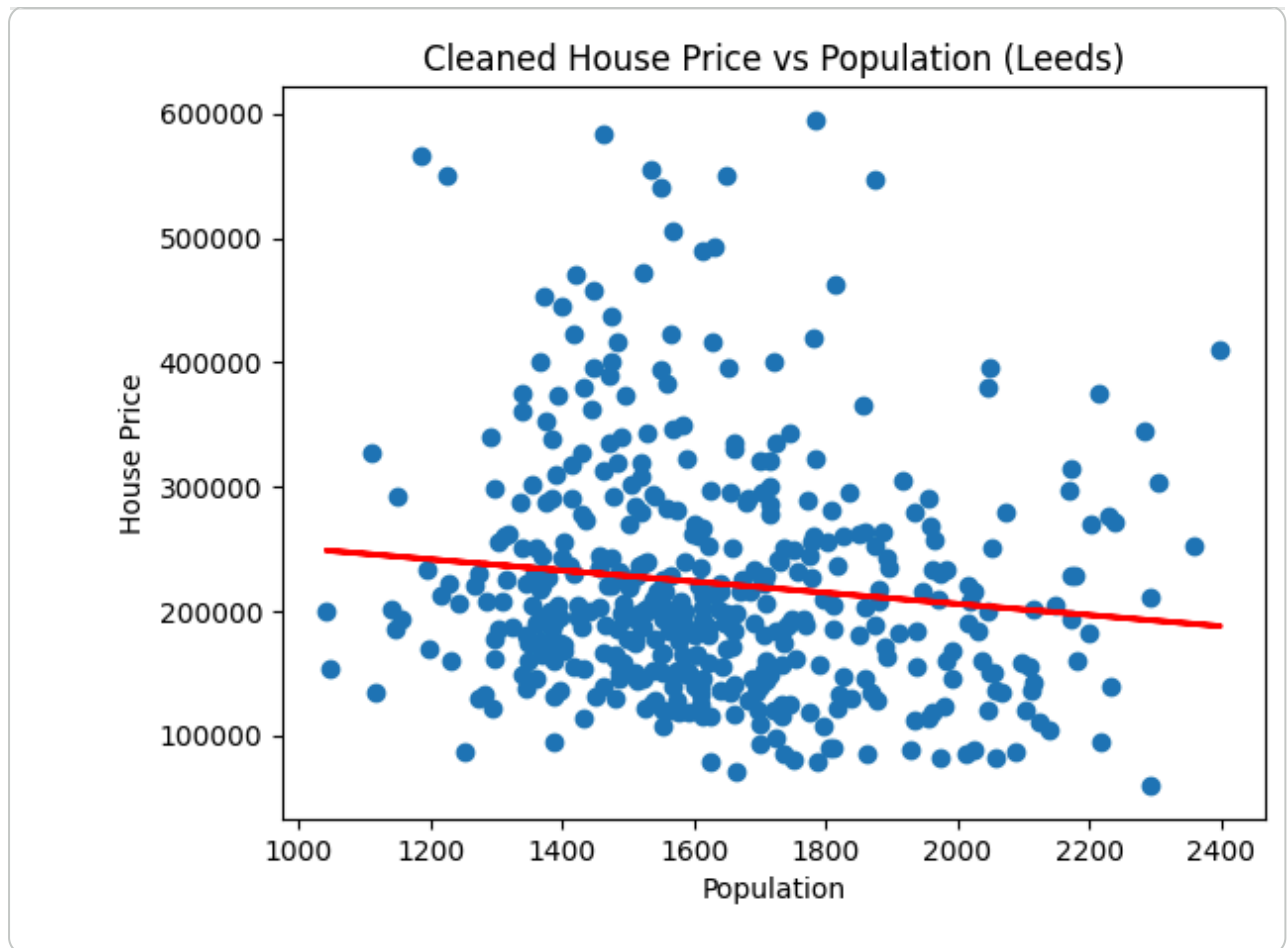
clean_df = merged_df[
    (merged_df["house_price"] < 600000) &
    (merged_df["population"] < 2500)
]
x = clean_df["population"]
y = clean_df["house_price"]

m, b = np.polyfit(x, y, 1)

plt.scatter(x, y)
plt.plot(x, m*x + b, color='red', linewidth=2)

plt.xlabel("Population")
plt.ylabel("House Price")
plt.title("Cleaned House Price vs Population (Leeds)")
plt.show()

```



✓ Exploratory Analysis

An initial analysis explored the relationship between population and house prices in Leeds. The scatter plot shows a weak correlation between the two variables, indicating that population alone is not a strong predictor of house prices.

Therefore, the focus of the analysis shifted to socio-economic deprivation (IMD), which is expected to have a stronger relationship with housing values.

IMD scores represent levels of socio-economic deprivation across areas in England [1].

```
imd_df = pd.read_csv('leeds_imd_2019.csv')
```

```
imd_df.head()
```

```
/tmp/ipykernel_17771/1319396865.py:1: DtypeWarning: Columns (0) have mixed types. Spec
imd_df = pd.read_csv('leeds_imd_2019.csv')
```

	LSOA_code	IMD_Score	
0	E01011264	20.525	
1	E01011265	13.602	
2	E01011266	5.808	
3	E01011267	27.863	
4	E01011268	34.444	

```
imd_df.columns = imd_df.columns.str.strip() #
imd_df = imd_df.rename(columns={
    'LSOA code (2011)': 'LSOA_code',
    'LSOA code': 'LSOA_code',
    'IMD Score': 'IMD_score',
    'Index of Multiple Deprivation (IMD) Score': 'IMD_score'
})
merged_df = price_df.merge(pop_df, on='LSOA_code')
merged_df = merged_df.merge(imd_df, on='LSOA_code')

merged_df.head()
```

	LSOA_code	house_price	area_name	population	IMD_Score	
0	E01011264	207500.0	Leeds 011A	1284.0	20.525	
1	E01011265	289950.0	Leeds 009A	1955.0	13.602	
2	E01011266	332500.0	Leeds 008A	2613.0	5.808	
3	E01011267	228000.0	Leeds 009B	1699.0	27.863	
4	E01011268	175000.0	Leeds 010A	1395.0	34.444	

Next steps:

[Generate code with merged_df](#)

[New interactive sheet](#)

```

import matplotlib.pyplot as plt
import numpy as np

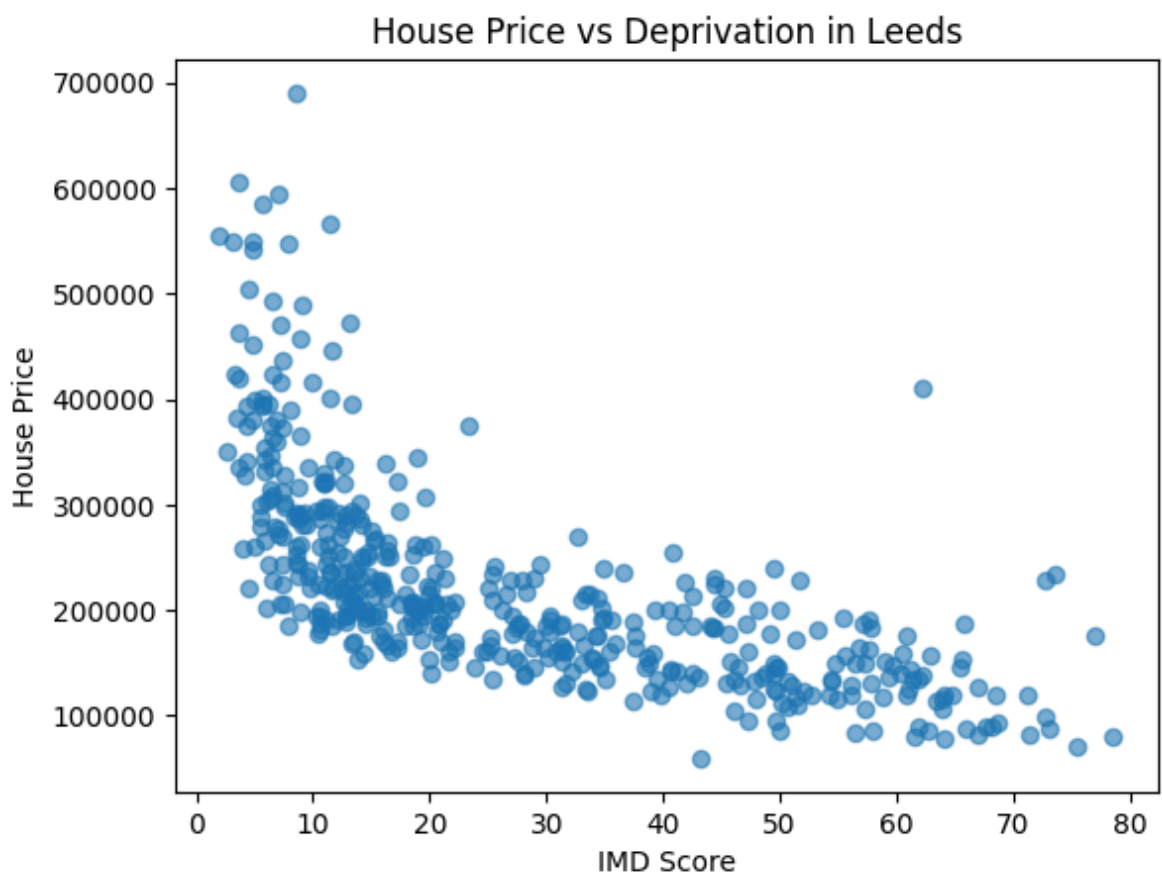
x = merged_df["IMD_Score"]
y = merged_df["house_price"]

m, b = np.polyfit(x, y, 1)

plt.scatter(x, y, alpha=0.6)
plt.plot(x, m*x + b, color="red", linewidth=2)

plt.xlabel("IMD Score")
plt.ylabel("House Price")
plt.title("House Price vs Deprivation in Leeds")
plt.show()

```



```

corr = merged_df["IMD_Score"].corr(merged_df["house_price"])
print("Correlation:", corr)

```

Correlation: -0.6638117682027144

✦ Relationship between IMD and House Price

The scatter plot shows a clear negative relationship between IMD score and house prices in Leeds.

Areas with higher deprivation (higher IMD scores) tend to have lower house prices. This is supported by the correlation coefficient ($r \approx -0.60$), indicating a moderately strong negative relationship.

This suggests that socio-economic conditions play an important role in determining property values.

The spatial boundary data (Leeds.geojson) was provided as part of the GEOG5990M course materials [2].

```
import geopandas as gpd
```

```
map_df = gpd.read_file("Leeds.geojson")
map_df.head()
```

	FID	LSOA21CD	LSOA21NM	BNG_E	BNG_N	LONG	LAT	GlobalID
0	10720	E01011264	Leeds 011A	420937	441836	-1.68306	53.87232	14a73329 fc8f-464 a4fa 1963baa3550
1	10721	E01011265	Leeds	418988	441951	-1.71269	53.87343	eda471dc f408-4da

Next steps:

[Generate code with map_df](#)

[New interactive sheet](#)

```
map_merged = map_df.merge(
    merged_df,
    left_on="LSOA21CD",
    right_on="LSOA_code"
)
```

```
map_merged.head()
```

	FID	LSOA21CD	LSOA21NM	BNG_E	BNG_N	LONG	LAT	GlobalID
0	10720	E01011264	Leeds 011A	420937	441836	-1.68306	53.87232	14a73329 fc8f-464 a4fa 1963baa3550
1	10721	E01011265	Leeds 009A	418988	441951	-1.71269	53.87343	eda471dc f408-4da 917e c5e188310b3
								286c32c -f40-464

Next steps:

[Generate code with map_merged](#)[New interactive sheet](#)

```

import matplotlib.pyplot as plt

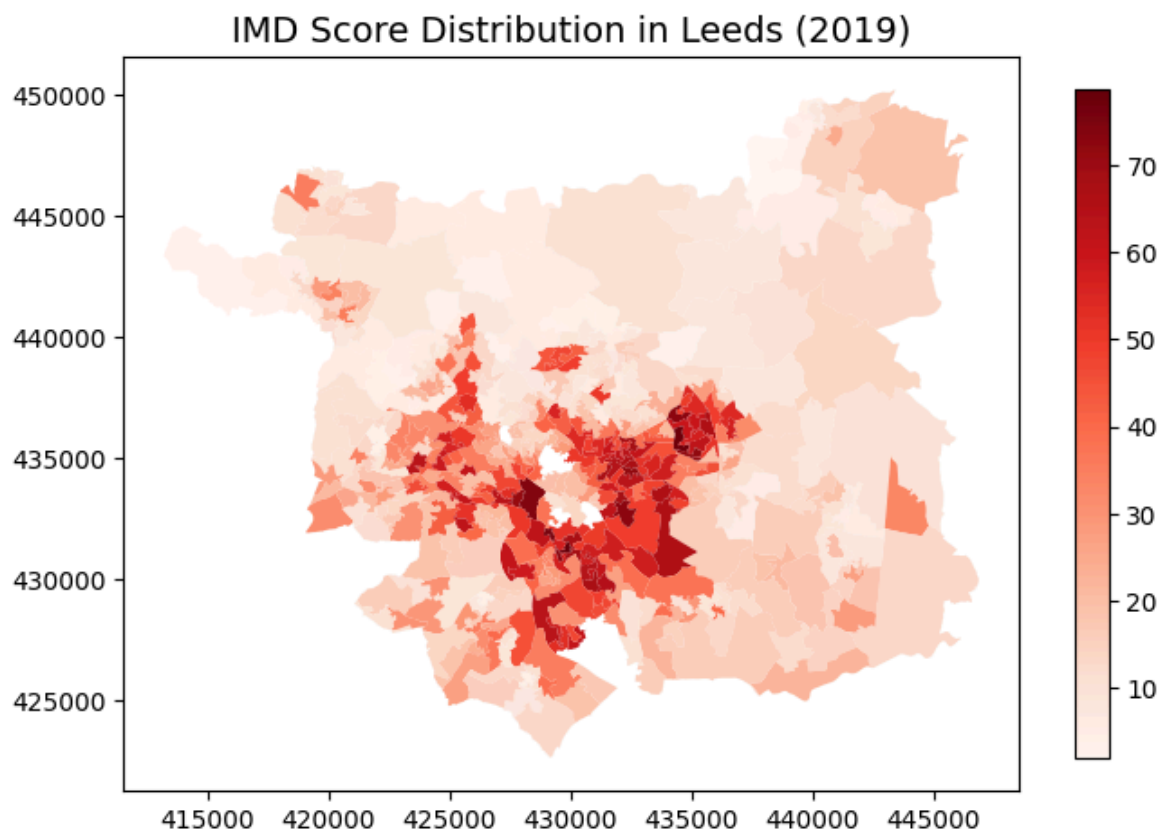
fig, ax = plt.subplots(1, 1, figsize=(8, 8))

map_merged.plot(
    column="IMD_Score",
    legend=True,
    cmap="Reds",
    ax=ax,
    legend_kwds={'shrink': 0.6}
)

ax.set_title("IMD Score Distribution in Leeds (2019)", fontsize=14)

plt.show()

```



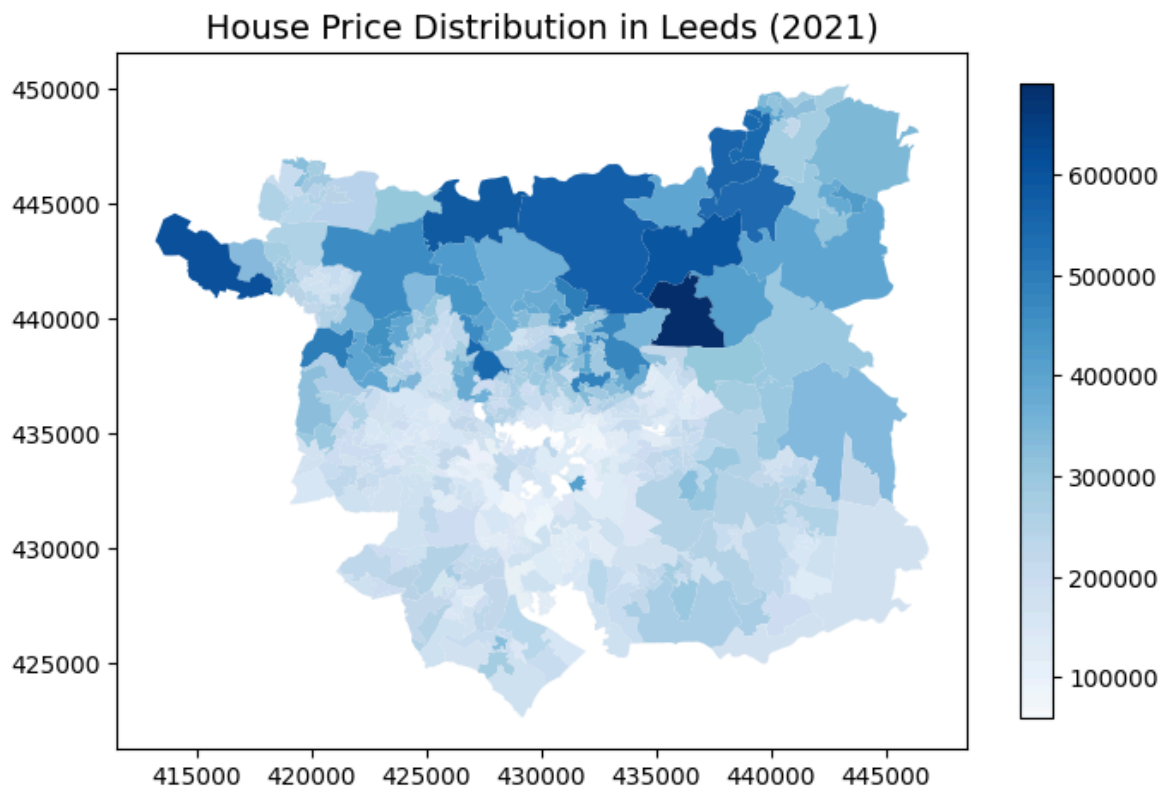
```

fig, ax = plt.subplots(1, 1, figsize=(8, 8))

map_merged.plot(
    column="house_price",
    legend=True,
    cmap="Blues",
    ax=ax,
    legend_kwds={'shrink': 0.6}
)

```

```
ax.set_title("House Price Distribution in Leeds (2021)", fontsize=14)  
  
plt.show()
```



Spatial Patterns

The spatial distribution shows clear geographic differences across Leeds.

Higher deprivation levels are concentrated in central areas, while higher house prices are found in less deprived outer areas.

This spatial pattern supports the negative relationship observed in the scatter plot